The Sectoral Impact of Brexit on UK–EU Goods Trade: A Difference-in-Differences Approach

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Abstract

This paper examines the short-term, sectoral impact of Brexit on UK goods exports to the EU27, utilizing a difference-in-differences approach with annual data from 2012 to 2023 at the HS2 commodity level. By comparing export trends to the EU27 and the rest of the world before and after the UK's withdrawal from the European Union, we identify the causal effects of Brexit. Our findings indicate that, up to the end of 2023, Brexit led to statistically significant declines in UK exports to the EU27 in the food and beverage sector and the clothing and textile sector—areas where the EU–UK Trade and Cooperation Agreement (TCA) introduced new non-tariff barriers. In contrast, most other sectors, including machinery, vehicles, gold, and oil, did not experience significant Brexit-related changes, with observed fluctuations attributable to global market factors such as the COVID-19 pandemic, the war in Ukraine, and shifts in international demand. The results are robust to alternative specifications and are cross-validated using both export and import datasets from UN Comtrade. This sectoral evidence underscores the importance of non-tariff barriers in shaping trade outcomes following major policy changes and provides actionable insights for policymakers negotiating future trade agreements.

Keywords: Brexit, UK exports, EU27, Non-tariff barriers, Sectoral analysis

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Introduction

Exports play a key role in the UK economy, directly influencing employment, income, and the country's economic resilience. The impact of Brexit on trade with the UK's largest trading partner, the EU27, has been a subject of significant interest not only among economists but also among policymakers, businesses, and the broader public. Brexit represents a major economic event, generating uncertainty, disputes, and various forecasts about the future of the UK economy. A particularly important issue is whether Brexit has adversely affected UK exports to the EU, and if so, which sectors have been most affected and the underlying causes of these changes.

This paper seeks to provide a clear and quantitative analysis of the short-term consequences of Brexit on UK goods exports to the EU27, in comparison with exports to the rest of the world. The study focuses on sectoral differences, aiming to identify which industries have been most impacted and what factors—such as new trade barriers, changes in demand, or global shocks—have contributed to these outcomes. To address this, a difference-in-differences approach is employed at the HS2 commodity level, using data from UN Comtrade covering the period from 2012 to 2023. Particular attention is given to the influence of the EU-UK Trade and Cooperation Agreement (TCA), which came into effect in 2021.

Prior research has focused on the effects of Brexit on either the overall economy or aggregated exports to the EU27. For instance, Dhingra et al. (2017, 2022) predicted a decline in welfare and exports if the UK left the single market, while Freeman et al. (2022), using a more differentiated approach, observed only a slight and temporary decrease in exports to the EU27. However, we have not found any studies that focus on a sectoral analysis with regard to the specific provisions of the TCA.

This paper makes the following contributions. It provides a sectoral assessment of export changes across 97 commodity categories, allowing for a deeper understanding of the sectoral differences and their relationship to specific provisions of the TCA. It compares export and import data to distinguish the effects of Brexit from global trends and other external shocks. It demonstrates that significant declines in exports to the EU27 are observed primarily in sectors where new non-tariff barriers were introduced after Brexit (e.g., phytosanitary requirements for food products and the lack of mutual recognition of textile standards), while in other sectors, changes are more likely attributable to global factors rather than Brexit itself.

The paper is organized as follows: The identification strategy and limitations of the difference-in-differences method are discussed in the next section. The following section presents the key results by sector, focusing on industries with the most significant export declines (e.g., food products and textiles), sectors with important export growth post-Brexit, as well as important sectors where no significant changes were observed or where growth was driven by global trends (e.g., gold, oil, aviation). The final section provides a discussion of the political and economic implications of the findings, along with suggestions for future research directions.

Literature review

Various authors have published research, both before and after Brexit, on the potential impact of Brexit on the UK economy. Several papers have focused on goods, and several have also used a difference-in-differences technique. Our paper contributes an analysis with a focus on HS2 categories of goods exports from the UK to the EU27 and specifically links the changes observed in exports to conditions imposed on trade in the post-Brexit EU-UK free trade agreement. Our paper doesn't focus on long-term effects, or economy-wide analysis, instead we concentrate on the immediate, industry-level effects of real-world trade agreements, in this case the TCA. This provides evidence to policymakers and trade negotiators, showing the immediate and tangible economic impacts of clauses and conditions in trade agreements.

Overall impact on exports (predictions pre-Brexit)

Before the TCA, in 2017, Dhingra et. al. estimated that a "soft-Brexit", where the UK remained in the single market, would have a negative effect on UK welfare of 1.3%, with the costs rising to up to 6.3% if flow-on effects of productivity are considered, and up to 9.4% in the case of a hard Brexit (Dhingra et al., 2017). Our analysis of the short-term impacts under a comprehensive free-trade agreement put us in the first scenario, where flow-on productivity effects are not considered and barriers to trade are minimised.

Dhingra and Sampson also found that there was a potential negative impact of Brexit on overall economic performance between 2016 and 2019, however there was no discernible anticipation effect for exports (Dhingra & Sampson, 2022). This supports our decision to analyse the effect of Brexit post 2021, rather than considering anticipatory effects.

Overall impact on goods exports (studies post Brexit)

Studies post-Brexit remain limited, as data has only recently become available. The studies that do exist showed similar findings to ours. Freeman 2022 used a difference-in differences method and estimate that there appears to have been only a small and temporary decline in UK goods exports to the EU27 following the implementation of the TCA (Freeman et al., 2022). The same authors also found again, that there was no significant anticipatory effect (Freeman et al., 2024). While Freeman used HS4 level data, they only showed results for aggregate exports, meaning they don't show the different impacts in each sector as we have done. Gasiorek also used a difference in differences model to find similar results (Gasiorek & Tamberi, 2023). However, they used a Poisson pseudo-maximum likelihood regression, which allowed them to consider fixed country effects. This is a level of detail which could have added further accuracy to our analysis.

Buigut used a gravity model to estimate changes in trade post-Brexit (Buigut & Kapar, 2023). They found significant drops in EU27 imports to the UK, but only a minor, temporary drop in exports. The use of a gravity model is interesting in that it allows for the inclusion of more explanatory variables than our simple difference-in-differences analysis.

Other factors that could affect exports

Other factors that could help explain our results have been examined by several authors:

Uncertainty

Uncertainty can also have an effect on exports and imports. Graziano 2023 estimates that Brexit, and the associated uncertainty are estimated to have caused inflation in the UK, which could have negative flow on effects for export competitiveness in the long-term (Graziano et al., 2023). The same author estimates that uncertainty pre-Brexit could negatively impact exports or imports, with exports decreasing with increasing uncertainty (Graziano et al., 2021). The signing of the EU-UK free trade agreement is likely to have significantly reduced uncertainty in the case of our analysis.

Impact on small firms

Smaller firms are predicted to be more impacted by any increased costs of exporting. Freeman et al 2022 show that there was a 30% fall in the variety of UK goods exported to the EU27 compared to the rest of the world post-Brexit, which was mostly offset by an increase in the volume of goods exported (Freeman et al., 2022). The authors interpret this as the impact of smaller exporters leaving the market. This could potentially explain some of the difference between sectors in our results, with both the food and textiles sectors, which showed declines in exports, having a relatively higher number of small firms.

Impact on other macroeconomic factors

Various authors have considered other possible effects of Brexit on the economy such as: Exadaktylos 2020 (Exadaktylos et al., 2020) who find that domestic firms become between 7-12% more productive after hiring foreign managers – with post-Brexit restrictions potentially affecting hiring of foreign talent; Zhu (2024) who finds that post-Brexit uncertainty has led to volatility and loss of value in financial markets; and Du 2022 (Du & Shepotylo, 2022) who estimates that Brexit may have reduced service exports to the EU27. These factors could have an indirect impact on goods exports, which we do not explore in this paper.

Data

Source

This paper shows the results of a difference in differences estimation for changes in exports of UK goods to the EU27 countries, compared to UK exports to the rest of the world, pre and post-Brexit.

The values used are the fob (free on board) value of exports from the UN Comtrade database. The data was taken at the HS2 commodity level, made up of 97 categories (99 classifications in total of which 2 are unused). The results at a country level were aggregated to create two variables for the regression, exports to the EU27 and exports to all other countries (ROW).

The period covered in the data is from 2012 to 2023, with data taken on an annual basis.

Following our first estimate, we conducted another difference-in-differences estimate for changes in imports of goods to the EU27 from the UK, compared to imports from the rest of the world. However, this estimation, using import data, is only used to verify the results from the first estimation, which is the principal estimation used in this paper. While conceptually, this second estimation represents imports to the EU27, the actual data used was the fob value of exports from the UK to the EU27 compared to the fob value of exports from all other countries to the EU27. This allows for a direct comparison of the two difference-in-differences estimates, without distortions due to the fact that Comtrade import data includes duties and other costs.

UK data reporting changed in 2021, which may have led to an increase in the value of exports included in data (Webb & Ward, 2024). This change has only affected the value of exports for small exporters, meaning it will not materially affect our findings. Firstly, because the main sectors of interest in our analysis showed significant drops in exports, a trend which would only be exacerbated in the absence of the change in data collection. Secondly, those sectors which showed significant increases in exports were dominated by large firms, for which the data reporting requirements have not changed.

Trade in services is not considered in this paper due to a lack of reliable data.

Descriptive statistics

Over the period 2012-2023, 45% of exports, by value, were to the EU27, while 55% were to the rest of the world. Machinery and electrical equipment represented 20% of total export value, vehicles and aircraft 14%, chemical products 13%, gold 12% and mineral products (oil and gas) 9.5%. Together, the 5 largest categories represented 70% of total exports from the UK, all destinations considered.

For machinery and electrical equipment, vehicles and aircraft, and chemical products around 50% of exports were destined for the EU27. For gold, only 13% of exports went to the EU27, while for oil and gas the figures are reversed, with 77% of exports going to the EU27.

For other smaller categories, the repartition of exports between the EU27 and the rest of the world changes significantly depending on the category. For example, for works of art and antiques, 9% went to the EU27, while for both food and textiles the EU27 was the destination for 67% of exports. Further details on the value of exports for each sector are included in table 6 in the appendix.

Identification strategy

Details of strategy

Here the aim is to identify the causal effect of Brexit on trade flows by using a differences-in-differences (diff-in-diff) strategy. The core idea is to compare the evolution of exports to EU countries versus non-EU countries before and after the United Kingdom's withdrawal from the European Union became effective (the period beginning in 2021). The following baseline equation is estimated separately for each product category:

$$\log(\text{fobvalue}_{it}) = \alpha + \beta_1 \cdot \text{EU}_i + \beta_2 \cdot \text{Post}_t + \beta_3 \cdot (\text{EU}_i \times \text{Post}_t) + \varepsilon_{it}$$

- fobvalue_{it} is the export value in FOB prices for partner i at time t,
- EU_i is a dummy variable equal to 1 if the partner country is in the European Union and 0 for the rest of the world,
- Post_t is also a dummy equal to 1 for years ≥ 2021 (post-Brexit), so for 2021, 2022 and 2023; otherwise it's 0 (from 2012 to 2020),
- $\text{EU}_i \times \text{Post}_t$ is the interaction term capturing the difference-in-differences,
- ε_{it} is the error term.

Here the coefficient of interest is β_3: it captures the differential change in exports to EU countries relative to non-EU countries from the pre-Brexit to the post-Brexit period. Under the parallel, or common, trend assumption, β_3 identifies the causal impact of Brexit on exports to the EU.

In other words:

- If $\beta_3 < 0$ and significant, **Brexit had a negative impact** on exports to the EU relative to the rest of the world.
- If $\beta_3 > 0$ and significant, **Brexit had a positive effect** on exports to the EU relative to the rest of the world (which may be counterintuitive and could suggest trade diversion or other effects).

Limitations of the identification strategy

For β_3 to truly capture the causal effect of Brexit, the model requires strong assumptions. The most critical of these is the **common trend assumption**, which states that, in the absence of Brexit, exports to EU and non-EU countries would have followed similar trends over time.

Under the common trend assumption and assuming no other event differentially affected trade with the EU at the same time, the interaction term $\text{EU}_i \times \text{Post}_t$ then isolates the Brexit effect by netting out:

- Time-invariant differences between EU and non-EU countries (through EU_i),

- Common shocks affecting all trade partners over time (through Post_t).

This design allows us to control for unobserved heterogeneity across countries and time, as long as it is not systematically correlated with the treatment (EU status) and the timing (post-2021).

However, several factors could threaten the validity of this identification strategy:

- **Violation of common trends:** If EU and non-EU trade flows were already evolving differently before 2021 due to other structural factors (e.g., regulatory harmonization, anticipation of Brexit effects, COVID-19), then the diff-in-diff estimate might capture more than just the Brexit effect.
- **Selection bias in treatment assignment:** Countries did not randomly become "treated" or "non-treated" — the classification is based on EU membership. If EU countries are systematically different in unobserved ways that evolve over time, this could confound the estimate.
- **Omission of relevant variables:** This occurs when key factors that influence the dependent variable (in this case, exports) and the assignment to a treatment group (EU vs. non-EU countries) are not included in the model. In the context of this Brexit analysis, such factors may include structural differences between countries or external shocks such as COVID-19.
- **Simultaneous causality:** This occurs when two variables simultaneously influence each other, creating a bidirectional causal loop. For instance, Brexit could influence exports, but exports may also influence trade policies. This can lead to biased coefficient estimates.
- **Measurement error:** This occurs when explanatory variables (e.g., exports) are measured incorrectly or imprecisely. For example, since we are measuring the value of exports, changes could reflect either price or quantity fluctuations — potentially driven by productivity, inflation, or trade volume variations.

What we can do to reinforce the credibility of the causal interpretation of β_3:

- **Trend analysis:** Check whether trends were parallel before 2021 (e.g., from 2012 to 2020). If the trends are similar prior to 2021, this supports the idea that divergence after 2021 is attributable to Brexit. This is particularly useful for detecting anticipation effects. Economic actors may have anticipated Brexit's impact, which could distort post-2021 causal estimates. Results for the common trend assumption are included for each sector in Table *6* in the appendix.
- **Placebo test:** Estimate the model using a false treatment date (e.g., pretending Brexit occurred in 2016 or 2018). If β_3 is insignificant for these placebo periods, it supports the robustness of the actual post-2021 effect. Note: A placebo test was not used here due to possible confounding events around these years, such as COVID-19 or other trade agreements.

- **Inclusion of new control variables or fixed effects:** Helps reduce the impact of omitted variables. New control variables or fixed effects were considered in this paper.
- **Use of an instrumental variable:** Could be used to address simultaneous causality. However, no IV was used here, as Brexit implementation is arguably exogenous to trade in goods.
- **Looking at similar industries:** Studying trends in closely related sub-industries (e.g., within the food sector or within the textile sector) can help determine if they were affected similarly. Consistent trends across similar subcategories support the hypothesis of a significant Brexit-related export change.
- **Comparing different datasets:** Addresses potential omitted variable bias. Two difference-in-differences estimations were conducted: one using UK exports to the EU27 vs. the rest of the world; the other using EU27 imports from the UK vs. imports from the rest of the world. If a relative change in UK exports was due to an unrelated factor (e.g. rising exports to the USA due to a change in US UK trade relations, or falling exports due to an overall decrease in European demand), the effect should appear only in the export dataset, while in the import dataset we would expect EU27 importing decisions to remain unchanged, or be the same for all countries. In our analysis we observe that for sectors where there was a significant decrease in exports, namely the food and textile sectors, there were similar changes in both datasets, suggesting an effect related to specific changes in UK-EU27 trade relations. On the other hand, sectors with increased exports post-Brexit do not show corresponding difference-in-differences effects in import data, suggesting other global factors were at play. This is discussed further in the results section.

Results

Overview

Up until the end of 2023, for the majority of goods, Brexit appears to have had no significant effect on the value of UK exports to the EU27 compared to exports to the rest of the world.

Exports in the Comtrade database are divided into 22 sectors, which are further divided into 97 categories of commodity. Of the 97 categories assessed, 62, representing 67% of total export value over the study period, did not see any significant change following Brexit.

The sectors where exports to the EU27 fell significantly compared to the rest of the world, represented 11% of total exports over the study period. Around one third of these exports were in the food and beverage sectors, one third in the clothing and textile sectors, and one third in other sectors. The falls in both these sectors can be attributed, at least in part, to new regulatory requirements for exports to the EU27 post-Brexit. This is discussed in more detail for each of the relevant sectors below.

Table 1: Sectors where exports to the EU27 fell post-Brexit, compared to exports to the rest of the world

Sector*	Number of HS2 sub-categories in sector	% of total exports in this sector	% of total exports in categories** with decline post-Brexit***	Range of DiD coefficients	Weighted average of DiD coefficients****
Food and beverage[a]	23 (12 declined)	6.4%	5%	[-0.19, -1.61]	-0.48
Textile and clothing[b]	21 (9 declined)	3.2%	2.3%	[-0.49, -1.25]	-1.09
Other sectors	8 categories declined	90.4%	5.6%	[-0.13, -0.87]	-0.26

[a]Includes animal products, vegetable products and food and beverage. [b]Includes leather and furs. 22 sectors/97 HS2 categories (*). Categories with statistically significant difference-in-differences coefficients at 5% level (**). Compared to exports to rest of the world (***). Only for categories with significant decreases (****). DiD = Difference-in-Differences (p¡0.05).

Sectors that saw a significant increase in exports to the EU27 post-Brexit, relative to exports to the rest of the world, accounted for 22% of exports over the study period and were made up of large industrial machinery, precision measuring and medical equipment and works of art. We were unable to find any evidence to suggest that Brexit related changes were responsible for these increases.

Table 2: Sectors where exports to the EU27 rose post-Brexit, compared to exports to the rest of the world

Sector	**% UK exports***	**DiD Coef.****	**HS2 Subcat.**
Nuclear reactors, boilers, machinery and mechanical appliances; parts thereof	16.6	0.21	1
Optical, photographic, cinematographic, measuring, checking, medical or surgical instruments and apparatus; parts and accessories	4.1	0.22	1
Works of art; collectors' pieces and antiques	0.9	2.62	1

*For the period 2012-2023 **All results significant at 5% threshold. DiD = Difference-in-Differences estimator.

For further details on all the sub-categories in these sectors, table 6 in the appendix shows the results of the difference-in-differences for all 97 export categories. Graphs showing the validity of the common-trend assumption for these sectors are also included in the appendix.

Categories where exports to the EU27 fell relative to the rest of the world post-Brexit

The two main sectors where UK exports to the EU27 fell post-Brexit, relative to UK exports to the rest of the world were the food and beverage sector and the textile and clothing sector.

For the remaining 8 categories where significant falls were recorded, we have not conducted an in-depth analysis. These categories were all small and heterogeneous. Some, such as tanning dyes or straw products, could be conceivably related to the textile and clothing sector, while others, such as toy manufacturing, have also been affected by new requirements in the TCA to provide proof of compliance with European standards - an issue which is discussed in more detail for the textile and clothing sector.

Food and beverage sector

Overall, the 23 categories in the food and beverage sector made up 6.4% of the value of goods exports from the UK during the study period. 67% of these exports were to the EU27. Post-Brexit, 12 of the 23 sub-categories in this sector, making up 5% of the total export value of goods from the UK, showed declines in exports to the EU27, compared to exports to the rest of the world. The difference in differences coefficients for these 12 sectors ranged from -0.19 to -1.61, with a weighted average of -0.48. For all 12 sectors, a significant decrease in exports to the EU27 were observed, along with a convincing common trend assumption. This suggests that, all else being equal, food and beverage exports the EU27 in these categories would have been 48% higher between 2021 and 2023 in the absence of Brexit. However, this figure should be interpreted with caution as there may still be other unobserved factors at play which have influenced changes in production and trade. Table 3 shows detailed results of the difference in differences estimation for all 23 categories. Graphs showing common trend assumptions are included in the appendix.

Post-Brexit, in May 2021, the UK and the EU27 implemented the EU-UK Trade and Cooperation Agreement (TCA) (European Union, 2021). This essentially amounts to a free trade

agreement for the movement of goods (Confederation of British Industry, 2021). However, one of the notable hurdles for goods exports in the TCA is new declarations and paperwork for food products of both plant and animal origin, mostly relating to phytosanitary requirements (House of Lords European Union Committee, 2021).

We hypothesize that these new, non-tariff barriers have caused the decline in food and beverage exports to the UK.

Another possible explanation of why the food and beverage sector may have been particularly affected in our data is the fact that food production and exports have a relatively short production cycle, meaning they may have reacted faster to the effects of Brexit than other industries, such as sectors in the manufacturing industry, where production and order lead times can sometimes span several years.

Table 3: Food and beverage: Difference in differences estimates for UK exports to the EU27 compared to UK exports to the rest of the world, pre and post 2021

HS2 Category	DID	% Change		Common trend	Sig.	R2	% Tot '23	% EU '23	EU Imp.
		ROW	EU						
Animals; live	-0.01	24	23			0.88	0.14	56	
Meat and edible meat offal	-1.05	78	-27	Strong	***	0.90	0.44	55	Y
Fish and crustaceans	0.00	-5	-4			0.92	0.38	68	
Dairy produce	-1.21	79	-42	Strong	***	0.82	0.47	32	Y
Animal originated products	-0.41	56	15			0.92	0.03	66	
Trees and plants	-0.48	8	-40	None	*	0.94	0.01	81	
Vegetables	-0.24	11	-13			0.76	0.10	55	
Fruit and nuts	-1.61	61	-100	Strong	***	0.98	0.03	64	Y
Coffee, tea	-0.20	-3	-23			0.88	0.08	60	
Cereals	0.60	-48	12			0.76	0.16	90	
Milling products	-0.48	21	-27	Moderate	***	0.65	0.09	38	Y
Oil seeds	-1.16	12	-105	None	**	0.90	0.03	69	
Vegetable saps	-0.01	13	12			0.65	0.03	56	
Vegetable materials	-0.68	20	-49			0.79	0.00	82	
Fats and oils	-0.54	65	11	Strong	***	0.98	0.16	74	Y
Meat/fish preparations	-0.77	40	-37	Strong	***	0.96	0.08	67	N
Sugars	-0.58	15	-43	Moderate	***	0.90	0.10	55	N
Cocoa	-0.40	38	-2	Strong	***	0.97	0.22	66	Y
Cereal preparations	-0.40	33	-7	Strong	***	0.94	0.52	55	Y
Vegetable preparations	-0.57	17	-40	Strong	***	0.96	0.10	62	Y
Edible preparations	-0.43	35	-8	Strong	**	0.92	0.57	59	Y
Beverages	-0.19	13	-6	Strong	*	0.91	2.07	35	Y
Food industry waste	-0.87	50	-37	None	***	0.71	0.29	42	

Note: DID = Diff-in-Diff. Significance: *** 0.1%, ** 1%, * 5%. % Tot '23 = % of total UK exports in this category in 2023. % EU '23 = % of exports in this category destined for the EU27 in 2023. "Y" indicates that the same, significant trend was found using EU27 import data. Percentages shown without % sign for compactness.

Further evidence that the declines in food and beverage exports observed here are caused by Brexit can be seen by comparing these results with the difference in differences estimates found using data on imports to the EU27 from the UK, versus imports to the EU27 from the rest of the world. Shown in the final column of table 3. Here we can see that the results are the same for 10 of the 12 categories where significant declines were observed in the original analysis. If a change in relative exports from the UK was due to a factor other than Brexit, for example, an unrelated increase in exports to the USA, we would expect to see a significant effect in the export dataset, but not in the import data. However, the significant declines observed in both datasets suggests that these changes were due to a factor in the relationship between the UK

and the EU27, not something that changed between the UK and the rest of the world or a change in the EU27 which affected imports from all destinations equally.

Textiles and clothing

UK exports of textiles, clothing and leather products to the EU27 fell post-Brexit, relative to exports to the rest of the world. Of the 21 categories of goods in these sectors, 9 showed significant declines, with a discernible common-trend assumption, while the rest showed no significant change. These relative declines in exports to the EU27 ranged from -49% to -125%, with a weighted average of -109%. The 9 categories showing a significant decline accounted for 72% of textile and leather product exports over the study period, and 2.3% of total UK exports. Graphs showing common trend assumptions are included in the appendix. Using the same logic employed above for the food and beverage sector, we can further confirm that the declines in textile, clothing and leather product exports observed here are caused by Brexit by comparing these results with the difference in differences estimates found using data on imports to the EU27 from the UK, versus imports to the EU27 from the rest of the world. Shown in the final column of table 4. Here we can see that the results are the same for all of the 9 categories where significant declines were observed in the original analysis. This suggests that the relative decline in exports to the EU27 was due to factors affecting trade between the UK and the EU27, rather than factors causing an increase in trade from the UK to the rest of the world or a decrease in EU27 imports from all destinations.

While this analysis suggests that in the absence of Brexit, textile, clothing and leather exports in these 9 categories to the EU27 would have been more than double what they were over the period 2021-2023, this result must be interpreted with caution as it may still be influenced by some factors unrelated to Brexit.

The sharp decline in EU-bound textile exports could be due to potential trade barriers, supply chain disruptions, or shifts in consumer demand affecting UK textile industries. A survey of UK fashion and textile companies showed that over 90% reported some increase in export costs due to Brexit (UK Fashion & Textile Association, 2021). Some analysts have pointed to increased burdens on exporters to provide evidence of compliance with EU standards, a new condition imposed in the TCA, or have pointed to shortages of labour for the textile industry due to restrictions on free movement of people post Brexit (Hemingray et al., 2023). Other analysts have hypothesized that the changes are due to re-exportation, with goods that previously entered the single market via the UK now using other ports (Du et al., 2024). While there appears to be little strong evidence for this last theory, the fact that imports to the EU27 from the rest of the world rose post-Brexit, while imports from the UK fell may offer some weak support.

Table 4: Textiles and clothing: Difference in differences estimates for UK exports to the EU27 compared to UK exports to the rest of the world, pre and post 2021

HS2 Category	DID	% Change ROW	% Change EU	Common trend	Sig.	R2	% Tot '23	% EU '23	EU Imp.
Raw hides and skins	-0.13	-38	-52			0.32	0.05	41	
Leather articles	-1.25	61	-64	Moderate	***	0.82	0.20	35	Y
Furskins and artificial fur	0.14	-89	-75			0.63	0.00	54	
Silk	0.38	-77	-39			0.70	0.00	75	
Wool and animal hair	0.15	-32	-17			0.55	0.08	57	
Cotton	-0.32	5	-28			0.60	0.04	38	
Vegetable textile fibres	-0.83	48	-35	Strong	***	0.81	0.01	31	Y
Man-made filaments	-1.17	63	-54	None	**	0.77	0.10	40	
Man-made staple fibres	-0.82	-8	-90	Moderate	*	0.44	0.05	30	Y
Wadding and nonwovens	0.05	2	7			0.35	0.08	52	
Carpets	-0.49	16	-33	Strong	**	0.92	0.06	56	Y
Special woven fabrics	-0.33	-15	-48			0.73	0.03	52	
Coated textile fabrics	-0.32	-4	-37			0.74	0.08	57	
Knitted fabrics	0.14	-19	-5			0.58	0.03	64	
Knitted apparel	-1.15	25	-90	Moderate	***	0.95	0.38	50	Y
Non-knitted apparel	-1.05	3	-102	Moderate	***	0.94	0.41	48	Y
Textile made-up articles	-0.22	-5	-26			0.44	0.18	46	
Footwear	-1.20	18	-102	Moderate	***	0.96	0.18	63	
Headgear	-0.60	0	-59	Moderate	***	0.86	0.03	51	Y
Umbrellas and walking-sticks	-0.79	9	-70	Moderate	***	0.91	0.00	51	Y
Feathers and artificial flowers	-0.25	28	2			0.97	0.01	75	Y

Note: DID = Diff-in-Diff. Significance: *** 0.1%, ** 1%, * 5%. % Tot '23 = % of total UK exports in this category in 2023. % EU '23 = % of exports in this category destined for the EU27 in 2023. "Y" indicates that the same, significant trend was found using EU27 import data. Percentages shown without % sign for compactness.

Categories where exports to the EU27 rose relative to the rest of the world post-Brexit:

Three categories of goods, out of 97, showed significant increases in UK exports to the EU27, relative to the rest of the world, post Brexit. However, none of these increases could be linked to Brexit, with most of them appearing to be a part of global trends. With the TCA ensuring that there were no added barriers to trade for these three sectors, these findings appear to confirm that the negative impact of Brexit on goods trade has so far been mostly limited to those sectors where the EU-UK free trade agreement imposed greater non-tariff barriers to trade.

Nuclear reactors, boilers, machinery and mechanical appliances

Changes in exports in this sector were mostly driven by increases in exports of aircraft engines to the EU27, potentially driven by Airbus production in Europe.

Among the commodities at HS2 classification level, the most important category to see a significant post-Brexit increase in exports to the EU27 relative to the rest of the world was

category 84, covering nuclear reactors, boilers, machinery and mechanical appliances. This sector made up 16.6% of total UK goods exports over the study period, and saw a relative increase post-Brexit, with a difference in differences coefficient of 0.22 and a strong common trend.

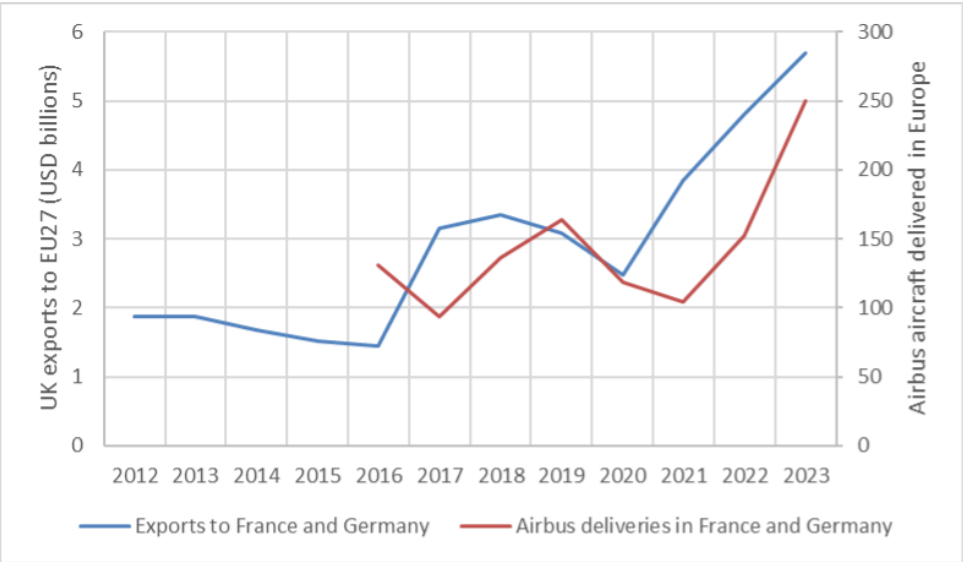
The relative change in exports to the EU27 was driven by an increase in export value to the EU27, while exports to the rest of the world remained stable. Looking at the three years pre-Brexit (2018-2020) compared to the three years after (2021-2023), average annual exports in this category to the rest of the world increased by 6%, while exports to the EU27 increased by 22%.

Further breaking down this category into its 95 HS4 subcategories reveals that among exports to the EU27, 96% of this change was due to increasing exports in category 8411, Turbo-jets, turbo-propellers and other gas turbines, which increased by 237%.

Meanwhile, the 6% increase in exports to the rest of the world was driven by increases in various sub-categories, with no single category having an overly large influence. The sub-category 8411 only recorded a 2% increase in exports to the rest of the world for this period.

Breaking category 8411 exports to the EU27 down further, to HS6 level, reveals that 92% of the increase in exports to the EU was due to aircraft engines. Rolls Royce is the principal manufacturer of aircraft engines in the UK, with a large share of their production purchased by Airbus, who produce aircraft in two locations in the EU27, France and Germany (Department for Exiting the European Union, 2017). Of the aircraft engine exports to the EU27 during the study period, 64% were destined for France and Germany. While data is not available for all years, making an econometric analysis difficult, a graphical comparison of the number of airbus delivered by the assembly lines in France and Germany, and the value of UK aircraft engine exports to the EU27 suggests that Airbus production could be a key driver of these exports.

Figure 1: UK aircraft engine exports



Note *Airbus production figure for 2019 is an estimate

Source: Airbus. (2025, March) Orders and deliveries: March 2025 summary.

Optical, measuring and medical equipment

HS2 category 90, "optical, photographic, cinematographic, measuring, checking, medical or surgical instruments and apparatus; parts and accessories" showed a significant increase in exports destined for the EU27, relative to the rest of the world post Brexit, with a strong common trend. Breaking this HS2 category down further to its 33 sub-categories at HS4 level showed that 4 of these subcategories displayed both a significant positive increase in exports to the EU27 and satisfied the common trend assumption.

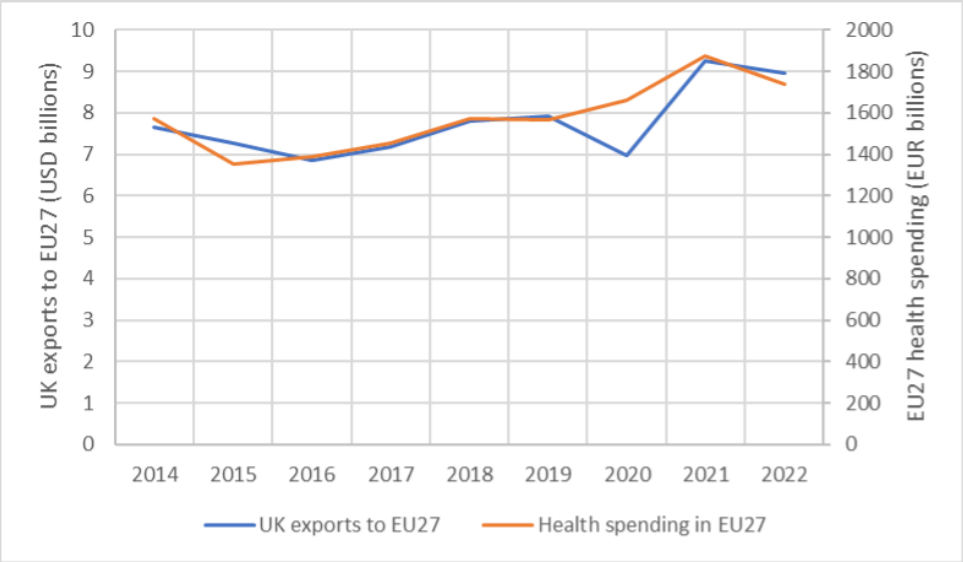
Table 5: Sub-categories of category 90 where significant changes in exports were observed

HS4 Sub-Category	Difference-in-Differences Coefficient	Share of Category 90 Export Value in This Category
Navigational instruments and appliances; direction finding compasses	1.1	5%
Instruments and appliances used in medical, surgical, dental or veterinary sciences, including scintigraphic apparatus, other electro-medical apparatus and sight testing instruments	0.27	17%
Apparatus based on the use of x-rays or of alpha, beta, gamma or other ionising radiations, whether or not for medical, surgical, dental or veterinary uses, including radiography or radiotherapy apparatus, x-ray tubes and other x-ray generators, high tension generators	0.59	1%
Instruments, apparatus for measuring, checking electrical quantities not meters of heading no. 9028; instruments, apparatus for measuring or detecting alpha, beta, gamma, x-ray, cosmic and other radiations	0.47	5%

*Note: For HS4 sub-categories accounting for 72% of export value in HS2 category 90, there was no significant change and/or common trend.

For the most important, sub-category here, medical instruments, we can likely explain this increase in exports post-Brexit as being related to EU27 health spending increases. Figure 2 shows the close relation between these two figures.

Figure 2: Health spending in the EU27 vs UK exports of medical instruments to the EU27 (scales adjusted)



Source: Eurostat. (2024) Current healthcare expenditure, 2014–22 (HCE2024)

Art

UK art and antiques exports to the EU27 rose significantly post-Brexit, relative to UK art exports to the rest of the world. There was also a strong common trend. However, it does not appear that this movement was related to Brexit.

The world art market is made up of 4 main players, the USA, China, the UK and the EU, representing 42%, 19%, 17% and 10 - 18% of the market respectively (McAndrew, 2025). Similar movements were seen in exports of art and antiques for all these regions over the study period, suggesting that global factors were at play, rather than Brexit. UBS analysis of the global art market suggests that large fluctuations in UK exports are mostly due to changes in demand in global markets, noting also that Brexit had not significantly changed conditions for art exports or imports to the UK (McAndrew, 2025). While this does not exclude effects of Brexit on art exports, it makes it difficult to separate any Brexit effect from larger, demand driven fluctuations.

Further, exports and imports of art do not reflect production, but rather art is often imported for sale at an auction house, before being re-exported to a final destination. This explains the fact that while export values have fluctuated over the study period, the share of the 4 major regions in sales in the art market has remained relatively steady. This further supports the case that despite a significant change in export values, there has been no significant effect of Brexit on the share of global art sold on the UK art market (McAndrew, 2024).

Figure 3: Worldwide art and antiques exports from major art markets

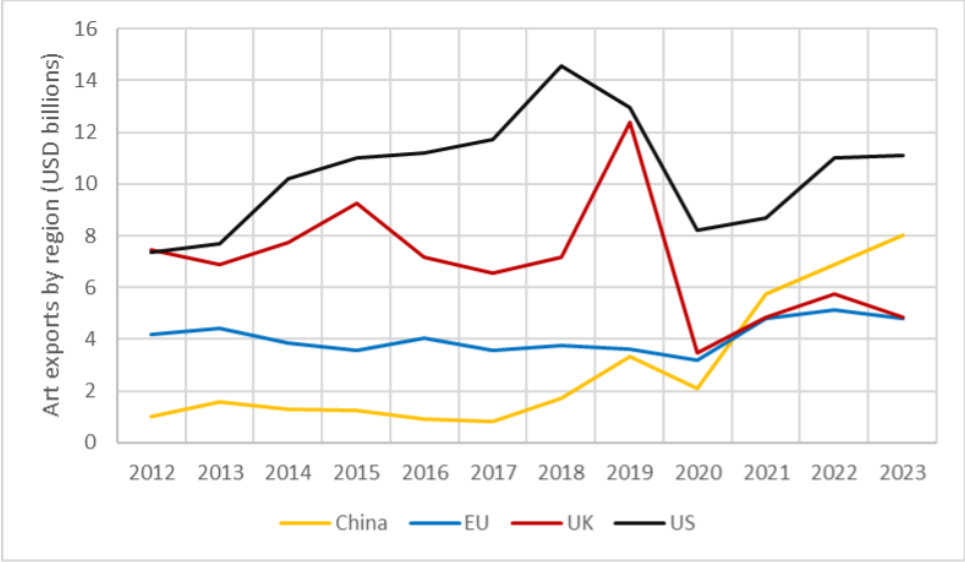
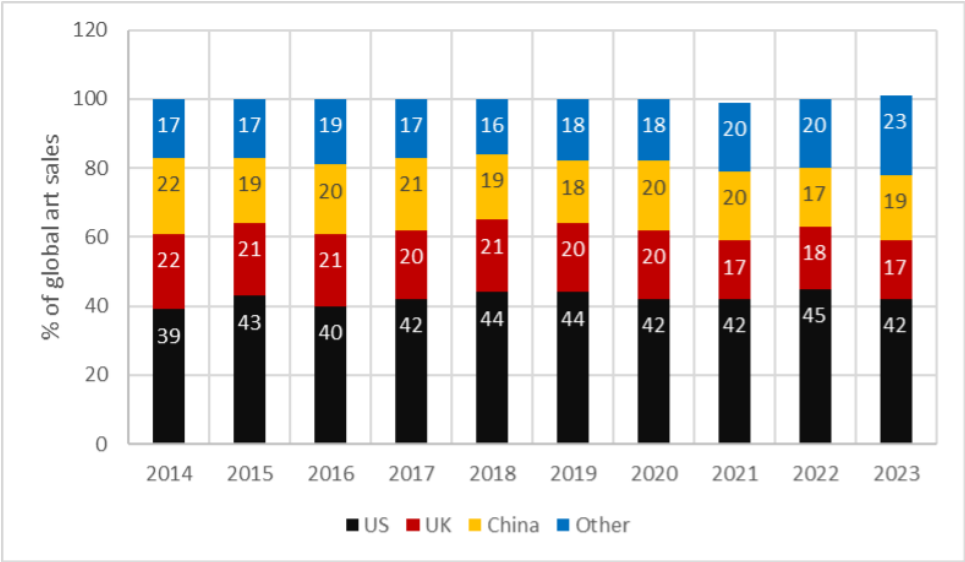


Figure 4: Global Art Market Share by Value of the US, the UK, and China 2014-2024



Source: (McAndrew, 2025)

Important export categories where no significant impact of Brexit was observed

The three largest categories of goods exports, by value, to not show a significant change in relative export levels to the EU27 post Brexit were pearls, precious metals and stones (essentially gold exports), mineral fuels and oils (essentially oil and gas), and vehicles. These categories accounted for 11.9%, 10.1% and 9.3% respectively of total UK export value during the study period. We have also included the aircraft category in this analysis, due to its important size (3.6%) and its potential relation to aircraft engine manufacturing, discussed in the previous sector.

While exports in these four categories did fluctuate over the study period, these changes can be attributed to global factors, unrelated to Brexit, such as COVID-19 disruptions to the

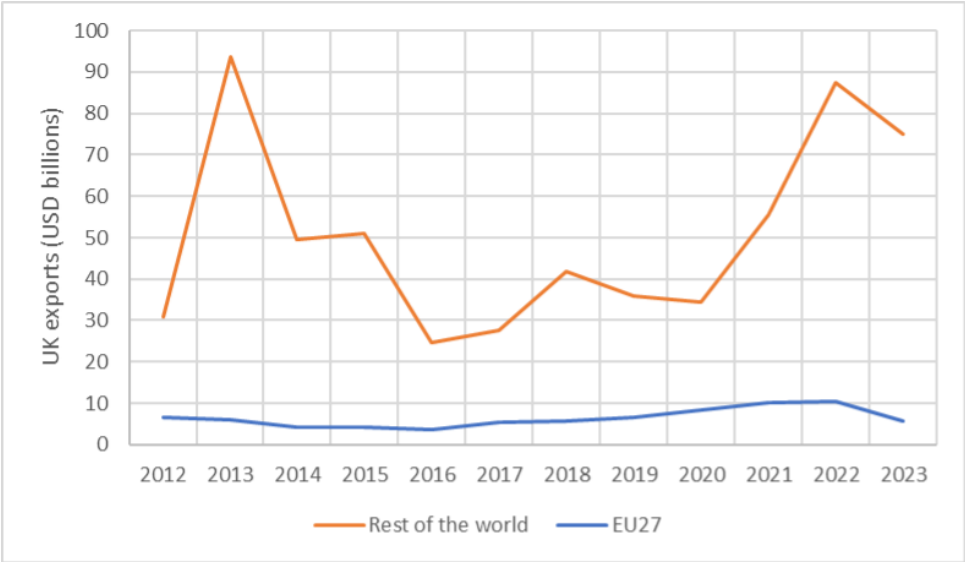
aircraft and vehicle industries, the effect of the war in Ukraine on oil and gas demand in Europe and the effect of global financial markets on demand for gold.

This once again confirms that, as of 2023, the sectors of UK goods exports to be negatively affected by Brexit are mostly those where the EU-UK free trade agreement introduced increased barriers to trade.

Gold exports

Since 2020, the export of gold and precious metals (code 71) from the UK has significantly increased, reaching 12% of the country's total goods exports in 2022 (ONS, 2023). Although one might expect this shift to be a consequence of the UK's departure from the European Union, available evidence suggests that the increase is primarily driven by global macroeconomic factors. In 2022, global demand for gold rose by 18% year-on-year, reaching 4,741 tonnes (World Gold Council, 2023). Central banks played a key role, increasing their gold reserves to a record level of 1,136 tonnes—the highest in the past 55 years. This surge in demand coincided with an increase in the average gold price to $1,800 per ounce, which elevated the export value, even though the physical volume remained relatively stable. Given that the UK, particularly London, serves as one of the world's major centres for gold storage and redistribution, the country was well positioned to meet growing international demand. Moreover, economic uncertainty triggered by the COVID-19 pandemic, rising inflation, and constrained global supply made gold an especially attractive asset. Therefore, the increase in UK gold exports after 2020 can be attributed less to Brexit and more to shifts in the international financial environment and the strengthening of the UK's role as a global gold trading hub (World Gold Council, 2023). This is further confirmed by the fact that the TCA does not introduce any additional barriers on the trade in gold (European Union, 2021).

Figure 5: UK exports of gold and precious metals to the EU27 and the rest of the world (2012–2023)



Oil and gas

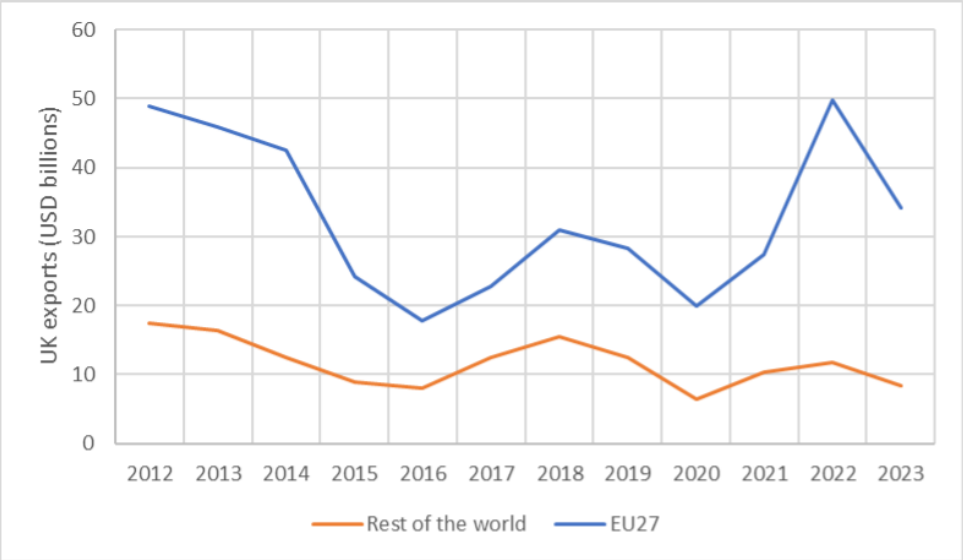
Since 2022, the United Kingdom has witnessed a significant increase in the export of oil and petroleum products (code 27), to both the EU27 and the rest of the world. This rise is closely tied to the geopolitical changes triggered by Russia's invasion of Ukraine and the subsequent European Union embargo on Russian oil, which came into effect at the end of 2022 and early 2023. As EU countries sought alternative suppliers, the United Kingdom capitalized on the situation and substantially increased its export volumes (IEA, 2023).

The average price of Brent crude oil also saw a sharp increase, rising from approximately $87 per barrel in January 2022 to a peak of around $123 in March, before decreasing to about $80 by December (Statista, 2023). Thus, even with the physical export volumes remaining constant, the surge in prices significantly boosted the overall export value.

While some EU countries—such as Bulgaria, Hungary, Slovakia, and the Czech Republic—continued to purchase Russian oil under temporary exemptions, the overall restructuring of European supply chains opened up new opportunities for the UK in the energy market (European Commission, 2023).

Therefore, the spike in British oil exports in 2022 can largely be explained by external factors—such as changes in demand, price trends, and the UK's strategic position in the post-sanctions energy landscape of Europe—rather than by shifts due to Brexit. Once again, the TCA did not impose any direct barriers to UK oil exports (European Union, 2021), further reinforcing the case that declines in trade were linked to barriers imposed in the TCA.

Figure 6: Export of oil and petroleum products from the UK to the EU27 and the rest of the world (2012–2023)



Vehicles and aircraft

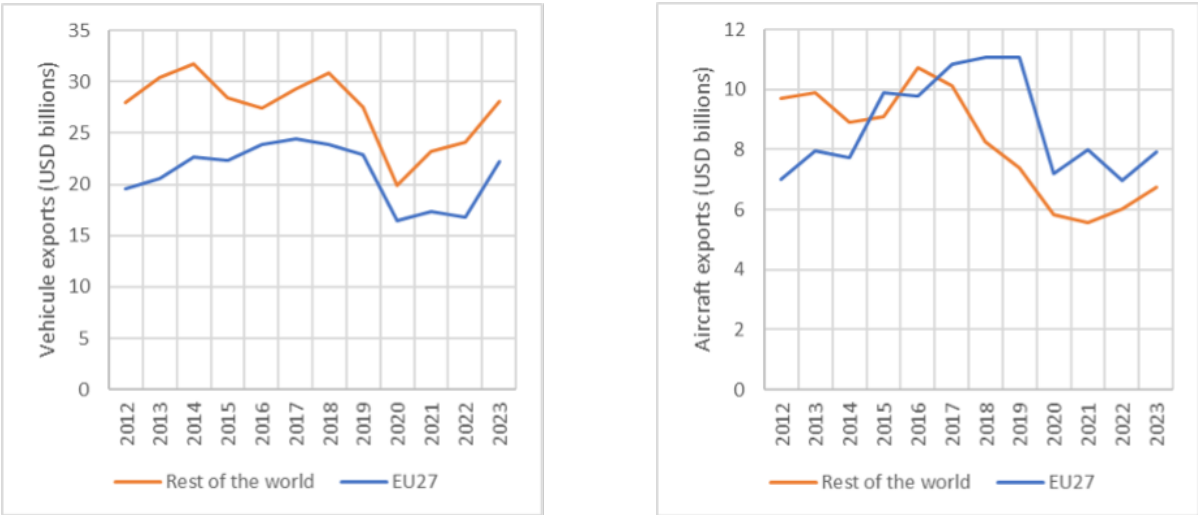
The vehicle and aircraft sectors are highly significant, contributing 10.14% to total UK exports during the study period. Exports of both vehicles and aircraft (excluding aircraft engines) suffered major declines from 2020, likely due to the COVID-19 pandemic and changing trade

conditions (Office for National Statistics, 2020).

While UK vehicle and aircraft exports to the EU27 experienced large declines post-Brexit, the difference-in-differences estimates were not significant, as exports to the rest of the world also fell over the same period. UK exports to the EU27 dropped by 15%, while exports to the rest of the world fell by 11% post 2020. This suggests that the change in exports was due to global factors, rather than Brexit.

Further, no barriers to vehicle trade were implemented in the TCA, with vehicle standards continuing to be regulated by international standards, no additional tariffs being introduced, and new rules of origin requirements not taking effect until 2024 (European Union, 2021). This provides further evidence that the sectors which suffered drops in exports post-Brexit were those for which the TCA imposed higher export barriers.

Figure 7: UK Aircraft and Vehicle Exports to the EU27 and the Rest of the World (2012–2023)



Verification of results using import data

There are competing narratives which could explain these results other than Brexit. First we can think of the COVID-19 crisis. This occurred around the same time, in 2020. Borders closed, there were serious disruptions to supply chains, along with supply and demand. It could be very difficult to dissociate the effects of COVID-19 from the impact of Brexit. Another possible explanation is that the relative change in exports was due to differences in trade with the rest of the world. For example, over this period, the UK may have begun exporting more to the USA, for reasons unrelated to Brexit. Finally, a drop in UK exports to the EU27 could have been due to factors in the EU27, such as an overall drop in EU27 demand for certain products.

To control for this we compared difference in differences estimations using two different datasets. One estimate used data on exports from the UK to the EU27 versus UK exports to the rest of the world. The other used data on imports to the EU27 from the UK versus imports from the rest of the world. If a change in relative exports from the UK was due to a factor other than Brexit, for example, an unrelated increase in exports to the USA, we would expect to see a significant effect in the export dataset, but not in the import data. For example, the change

in UK exports to the USA should not change sourcing decisions for imports to the EU27. In the case of global effects, such as for COVID-19, we would not expect to see significant effects for either dataset.

In effect, this is what can be seen in the analysis. The two sectors where exports are assumed to have declined due to Brexit, food and textiles, show the same movements in both datasets. For the food and beverage sector, 10 of the 12 categories that saw significant declines when using export data also saw declines when using import data. In the clothing and textiles sector, 10 of the 11 categories that saw declines when using export data also saw declines when using import data. There were no categories that recorded significant increases in the food or textiles sectors for either of the datasets.

For the two major sectors with increased exports post-Brexit—nuclear reactors, boilers and machinery, and optical, measuring, and medical equipment—this trend was not mirrored in import data, suggesting factors unrelated to Brexit. This aligns with our analysis of the post-Brexit EU27-UK free-trade agreement's impacts, with those sectors that saw increased barriers to trade in the TCA being the same sectors that saw a decrease in exports to the EU27.

Conclusion

Post-Brexit, the food and beverage and the textile and clothing sectors saw declines in UK exports to the EU27 relative to the rest of the world. These declines appear to be linked directly to non-tariff trade barriers introduced in the TCA. For other major sectors, where there were no direct trade barriers imposed in the TCA, non-Brexit related factors caused large changes. Gold exports increased due to changes in the global financial market, oil and gas exports increased following the war in Ukraine, COVID-19 reduced manufacturing exports, while aerospace exports were helped by increased production at Airbus. This sectoral analysis shows the importance of eliminating all barriers to trade, including non-tariff barriers, to fully benefit from free-trade agreements.

However, there are certain limitations regarding our choice of data. We relied on Comtrade data, which offers advantages such as accessibility and ease of use. However, it comes with several important drawbacks. One key limitation is the lack of granularity at HS2 level, as the data are aggregated at the level of product categories and countries, thereby restricting the depth of our analysis. Breaking down our analysis to a country or regional level and looking at finer scale categories could offer more insight into the drivers of export changes. Furthermore, the dataset is not consistently updated and may contain gaps for certain time periods, such as missing data for specific years or months. Given that Comtrade relies on self-reported declarations by individual countries, the data may be provisional or biased.

Our analysis is also limited in that it only shows the short-term, direct effects of Brexit over the last 3 years. Longer-term impacts on the movements of skilled labor, impacts on financial markets, inflation or exchange rates are likely to impact exports in the longer term. Nor does our analysis cover services, which constitute a substantial portion of trade between the United Kingdom and the European Union. As a result, while our study aligns with its title, it only partially captures the broader impact of Brexit.

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Appendix

Table 6: Difference in differences results for HS2 commodity categories - UK exports to the EU27 vs UK exports to the rest of the world.

HS2	Category	DID	%ΔROW	%ΔEU	Com.Trend	Sig.	R^2	%Tot	%EU	%ROW
1	Animals; live	−0.01	24	23			0.88	0.14	56	44
2	Meat	−1.05	78	-27	Strong	***	0.90	0.44	55	45
3	Fish	0.00	−5	-4			0.92	0.38	68	32
4	Dairy	−1.21	79	-42	Strong	***	0.82	0.47	32	68
5	Animal products n.e.s.	−0.41	56	15			0.92	0.03	66	34
6	Live plants	−0.48	8	-40	No	*	0.94	0.01	81	19
7	Vegetables	−0.24	11	-13			0.76	0.10	55	45
8	Fruit, nuts	−1.61	61	-100	Strong	***	0.98	0.03	64	36
9	Coffee, tea	−0.20	−3	-23			0.88	0.08	60	40
10	Cereals	0.60	−48	12			0.76	0.16	90	10
11	Milling products	−0.48	21	-27	Fair	***	0.65	0.09	38	62
11										
12	Oil seeds	−1.16	12	-105	No	**	0.90	0.03	69	31
13	Vegetable saps	−0.01	13	12			0.65	0.03	56	44
14	Vegetable materials	−0.68	20	-49			0.79	0.00	82	18
15	Fats, oils	−0.54	65	11	Strong	***	0.98	0.16	74	26
16	Meat/fish prep	−0.77	40	-37	Strong	***	0.96	0.08	67	33
17	Sugars	−0.58	15	-43	Fair	***	0.90	0.10	55	45
18	Cocoa	−0.40	38	-2	Strong	***	0.97	0.22	66	34
19	Cereal prep	−0.40	33	-7	Strong	***	0.94	0.52	55	45
20	Vegetable prep	−0.57	17	-40	Strong	***	0.96	0.10	62	38
21	Edible prep	−0.43	35	-8	Strong	**	0.92	0.57	59	41
22	Beverages	−0.19	13	-6	Strong	*	0.91	2.07	35	65
23	Food waste	−0.87	50	-37	No	***	0.71	0.29	42	58
24	Tobacco	−0.10	−135	-145			0.22	0.03	65	35
25	Salt, sulphur	−0.34	30	-5	Strong	**	0.91	0.13	54	46
26	Ores, slag	0.92	−53	39	No	***	0.87	0.03	45	55
27	Mineral fuels	0.35	−14	20			0.73	8.19	80	20
28	Inorg. chemicals	−0.44	83	39			0.40	0.69	48	52
29	Org. chemicals	0.20	−4	16			0.08	2.71	54	46
30	Pharmaceuticals	−0.16	−7	-23			0.43	5.28	38	62
31	Fertilizers	−0.27	63	36			0.57	0.09	52	48
32	Dyes, pigments	−0.20	9	-11	Strong	**	0.97	0.61	62	38
33	Essential oils	0.04	−2	2			0.95	1.11	66	34
34	Soap, detergents	−0.09	6	-2			0.96	0.45	63	37
35	Albuminoids	−0.10	39	29			0.88	0.15	55	45

Continued on next page

Table 6 – Continued from previous page

HS2	Category	DID	%ΔROW	%ΔEU	Com.Trend	Sig.	R^2	%Tot	%EU	%ROW
36	Explosives	−0.27	42	16			0.10	0.02	52	48
37	Photo goods	−0.29	−73	-102			0.53	0.03	69	31
38	Chemicals n.e.c.	−0.12	17	5			0.62	1.45	51	49
39	Plastics	−0.13	15	1	Strong	*	0.97	2.24	62	38
40	Rubber	−0.19	−2	-21			0.79	0.50	59	41
41	Raw hides	−0.13	−38	-52			0.32	0.05	41	59
42	Leather articles	−1.25	61	-64	Fair	***	0.82	0.20	35	65
43	Furskins	0.14	−89	-75			0.63	0.00	54	46
44	Wood articles	−0.31	62	31			0.94	0.15	66	34
45	Cork articles	−0.24	30	6			0.33	0.00	43	57
46	Basketware	−1.33	94	-39	No	**	0.44	0.00	47	53
47	Paper pulp	0.20	11	31			0.93	0.13	14	86
48	Paperboard	−0.08	−4	-12			0.95	0.53	65	35
49	Printed books	−0.28	−14	-42	Strong	*	0.71	0.56	41	59
50	Silk	0.38	−77	-39			0.70	0.00	75	25
51	Wool	0.15	−32	-17			0.55	0.08	57	43
52	Cotton	−0.32	5	-28			0.60	0.04	38	62
53	Veg. fibres	−0.83	48	-35	Strong	***	0.81	0.01	31	69
54	Man-made filaments	−1.17	63	-54	No	**	0.77	0.10	40	60
55	Man-made fibres	−0.82	−8	-90	Fair	*	0.44	0.05	30	70
56	Nonwovens	0.05	2	7			0.35	0.08	52	48
57	Carpets	−0.49	16	-33	Strong	**	0.92	0.06	56	44
58	Special fabrics	−0.33	−15	-48			0.73	0.03	52	48
59	Coated fabrics	−0.32	−4	-37			0.74	0.08	57	43
60	Knitted fabrics	0.14	−19	-5			0.58	0.03	64	36
61	Knitted apparel	−1.15	25	-90	Fair	***	0.95	0.38	50	50
62	Woven apparel	−1.05	3	-102	Fair	***	0.94	0.41	48	52
63	Textile articles	−0.22	−5	-26			0.44	0.18	46	54
64	Footwear	−1.20	18	-102	Fair	***	0.96	0.18	63	37
65	Headgear	−0.60	0	-59	Fair	***	0.86	0.03	51	49
66	Umbrellas	−0.79	9	-70	Fair	***	0.91	0.00	51	49
67	Feather articles	−0.25	28	2			0.97	0.01	75	25
68	Stone articles	−0.11	8	-3			0.83	0.23	55	45
69	Ceramics	−0.13	1	-12			0.16	0.11	47	53
70	Glassware	−0.49	39	-10	Strong	***	0.94	0.28	51	49
71	Precious metals	−0.14	58	44			0.92	15.50	7	93
72	Iron, steel	−0.12	18	6			0.07	1.50	47	53

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Table 6 – Continued from previous page

HS2	Category	DID	%ΔROW	%ΔEU	Com.Trend	Sig.	R^2	%Tot	%EU	%ROW
73	Iron articles	0.26	−4	22	No	*	0.25	1.30	53	47
74	Copper articles	−0.17	25	8			0.20	0.41	41	59
75	Nickel articles	−0.09	41	31			0.52	0.45	46	54
76	Aluminium articles	−0.32	35	3	Strong	**	0.86	0.64	51	49
78	Lead articles	0.36	−13	23			0.77	0.12	79	21
79	Zinc articles	−0.40	−1	-41			0.64	0.02	52	48
80	Tin articles	−0.51	−7	-58			0.29	0.00	51	49
81	Other metals	−0.01	23	22			0.40	0.26	56	44
82	Metal tools	−0.12	−17	-30			0.48	0.16	50	50
83	Misc metal products	−0.18	11	-6			0.60	0.26	49	51
84	Machinery	0.21	5	26	Fair	*	0.92	16.61	38	62
85	Electrical equip.	−0.04	−3	-7			0.11	5.46	49	51
86	Railway equip.	0.05	43	48			0.66	0.13	30	70
87	Vehicles	−0.04	−11	-15			0.59	9.66	44	56
88	Aircraft	0.19	−36	-17			0.40	2.81	54	46
89	Ships	0.12	−7	5			0.44	0.31	38	62
90	Optical instruments	0.22	3	25	Strong	***	0.94	4.13	44	56
91	Clocks	−0.34	18	-16			0.92	0.19	19	81
92	Musical instruments	−0.18	32	13			0.51	0.03	59	41
93	Arms	0.41	68	108			0.47	0.31	40	60
94	Furniture	0.05	−3	2			0.04	0.76	46	54
95	Toys	−0.68	37	-31	Strong	***	0.95	0.40	57	43
96	Misc manufactures	−0.87	18	-69	Strong	***	0.98	0.09	59	41
97	Artworks	2.62	−66	195	Strong	***	0.98	0.93	29	71
99	Unspecified	−2.30	94	-136	No	***	0.82	3.76	1	99

Note: HS2 category names have been truncated for simplicity. For the full names, please see `https://unstats.un.org/unsd/classifications/Econ`

Charts for analyzing common trend assumption for categories with significant changes in exports post-Brexit

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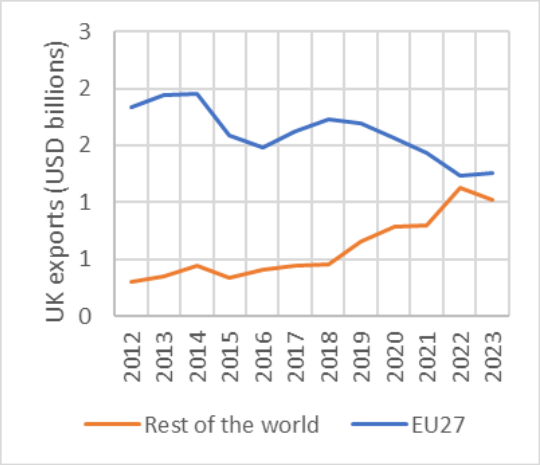

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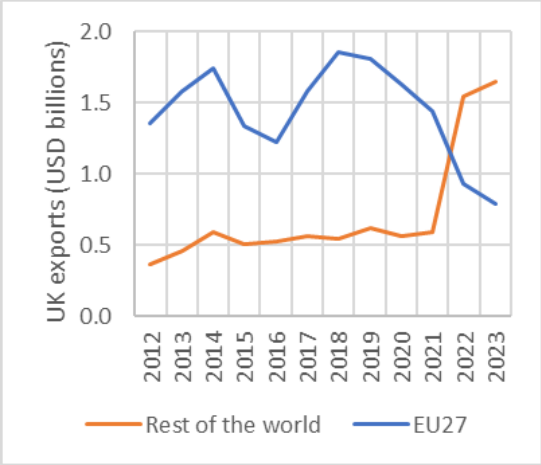

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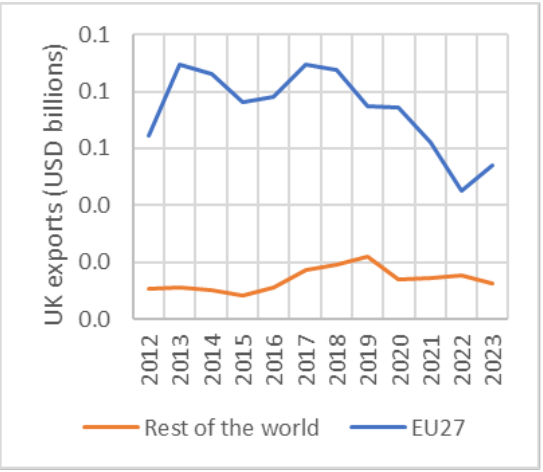

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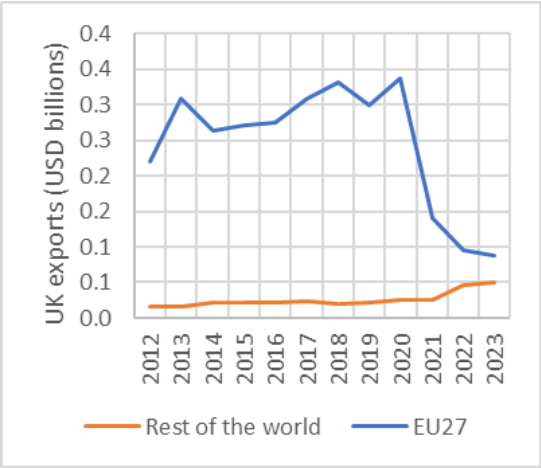

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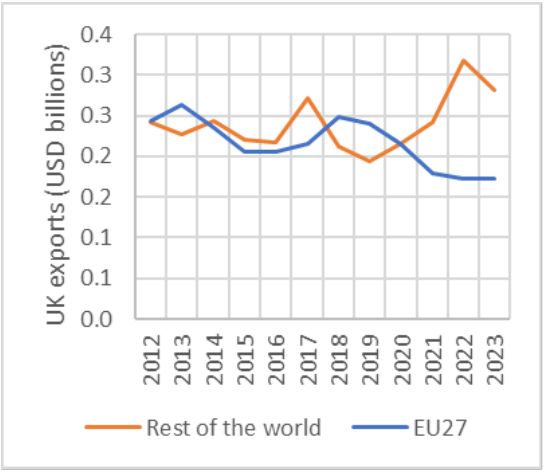

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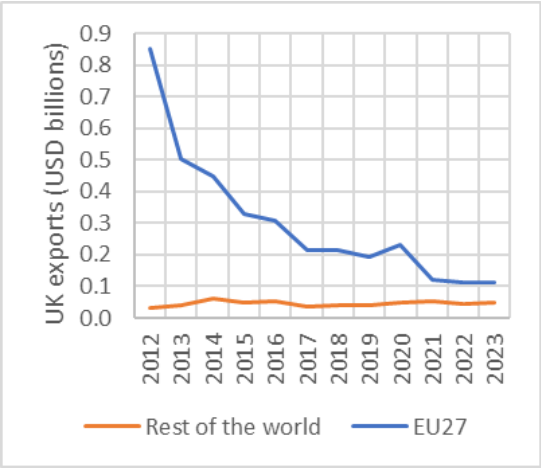

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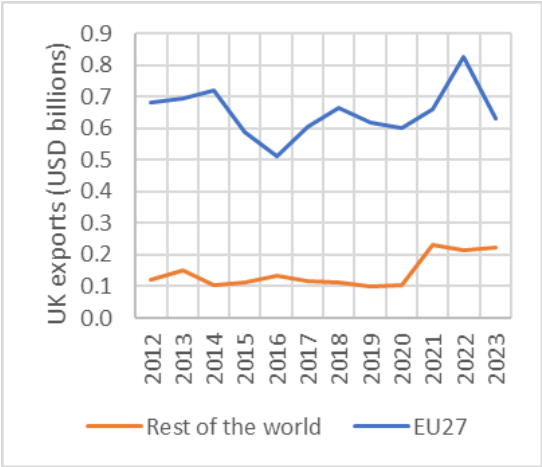

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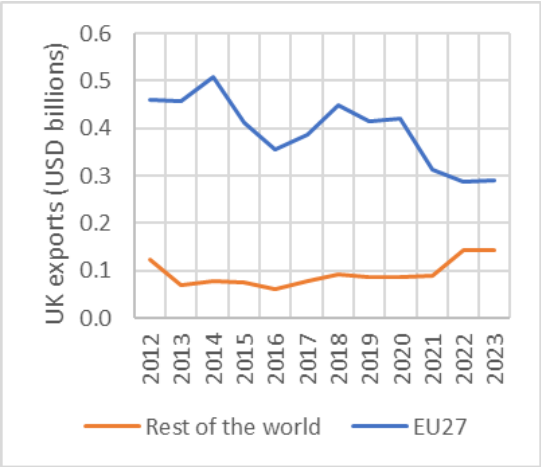

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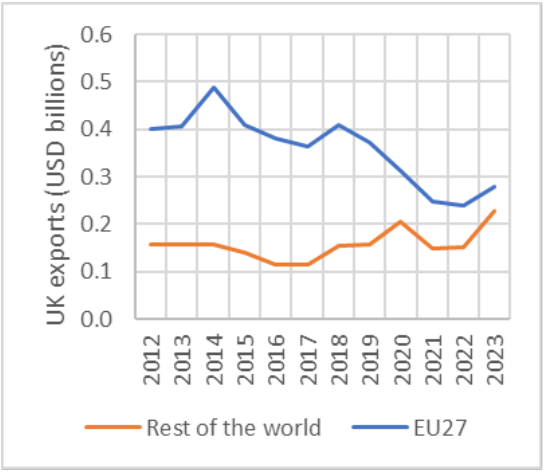

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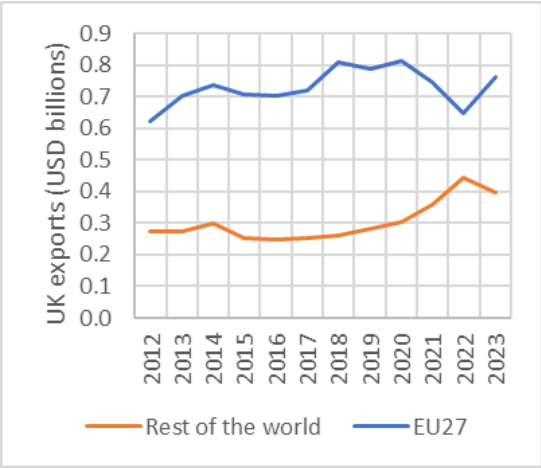

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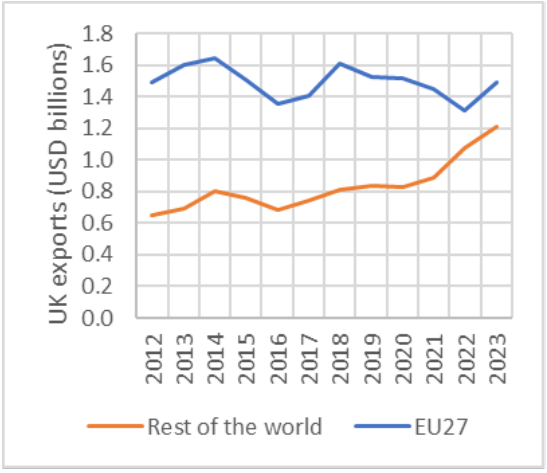

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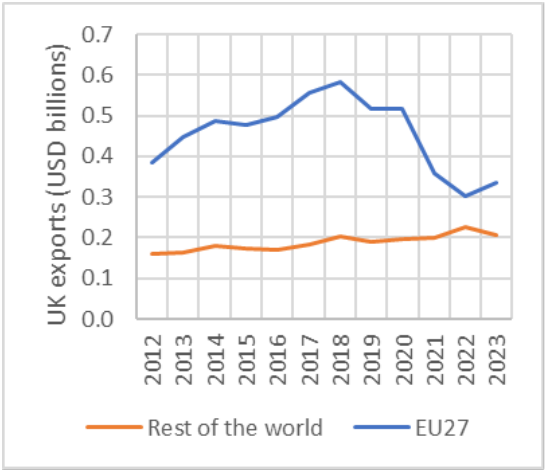

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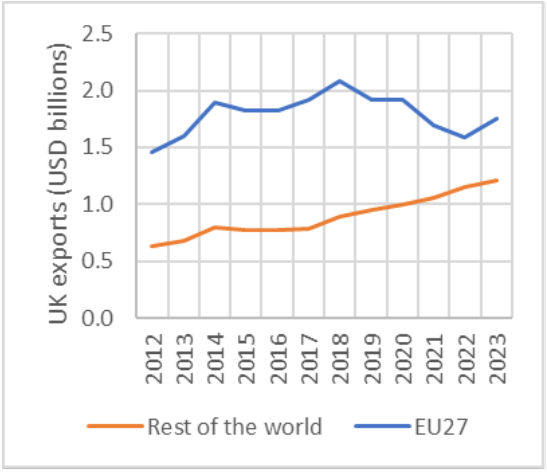

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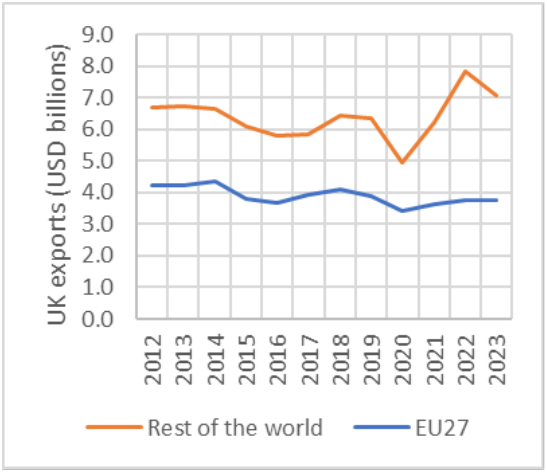

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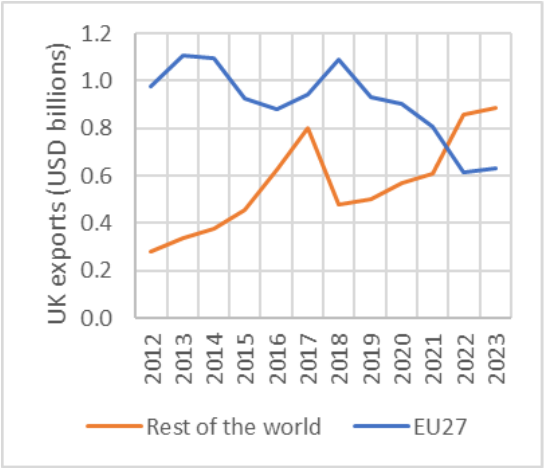

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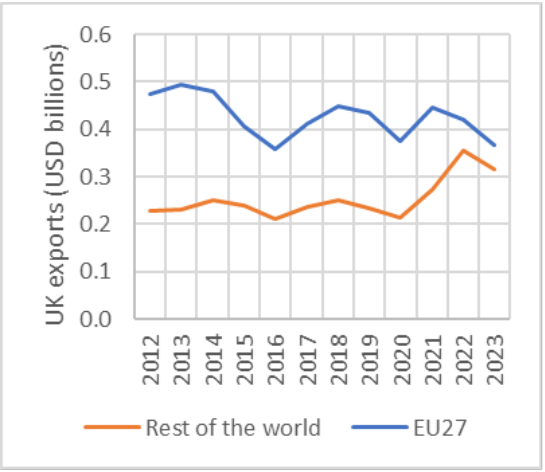

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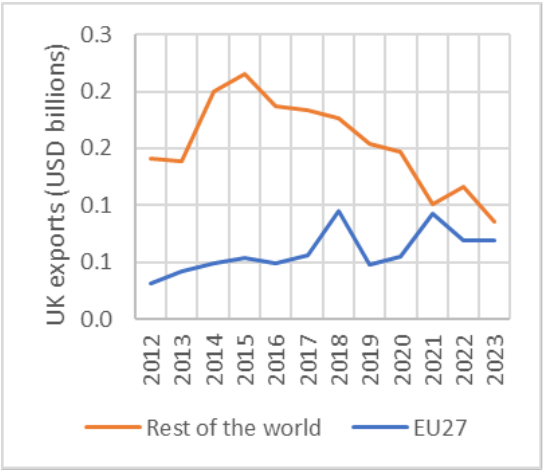

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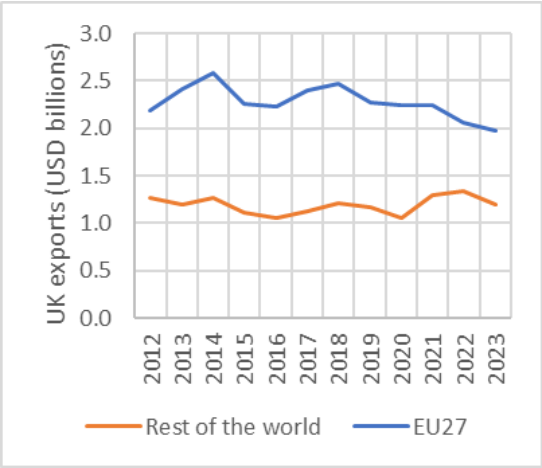

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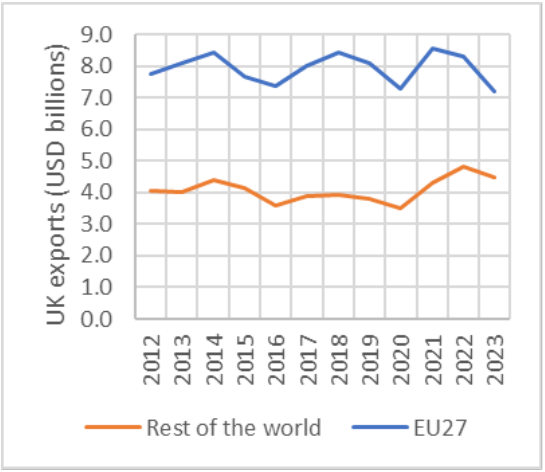

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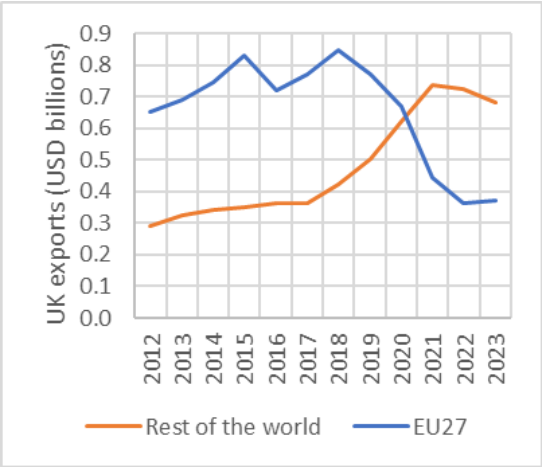

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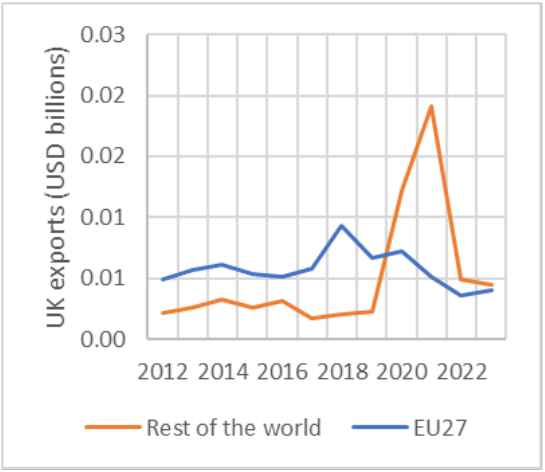

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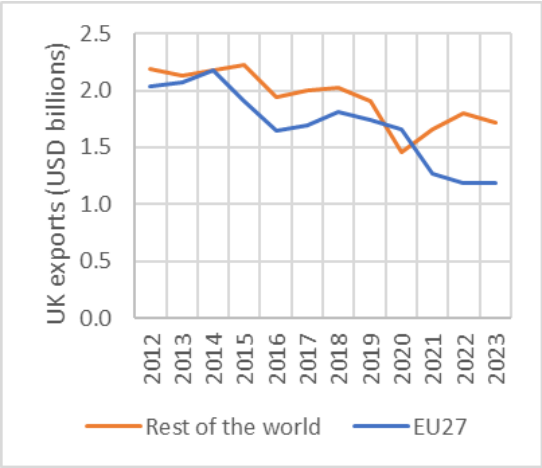

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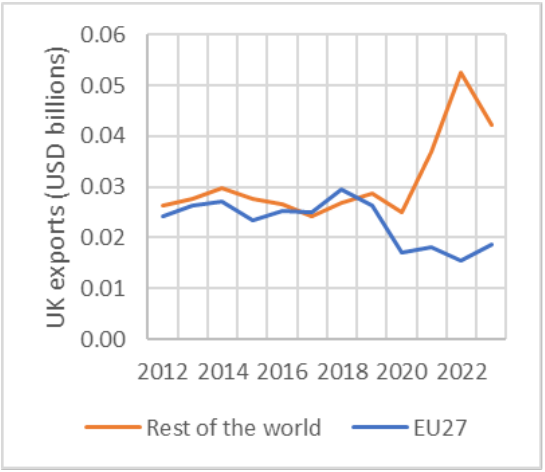

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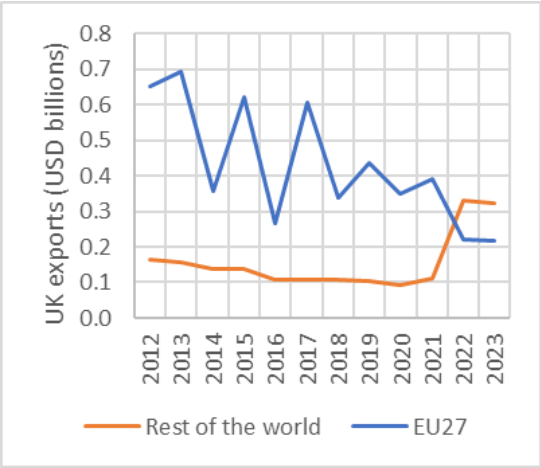

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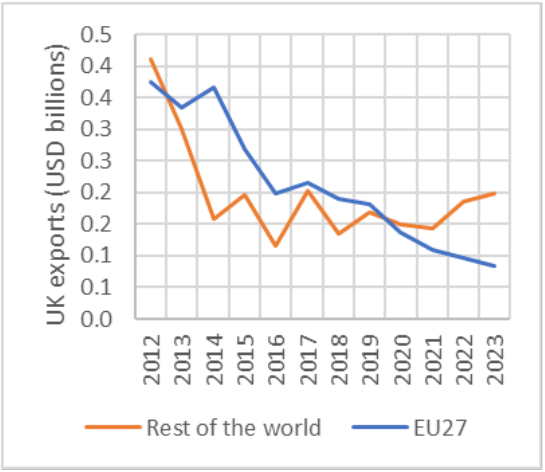

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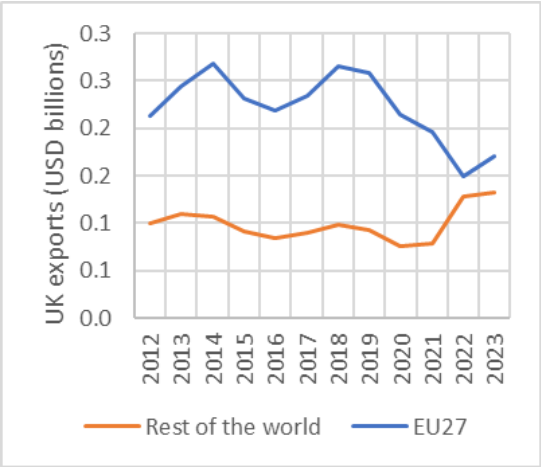

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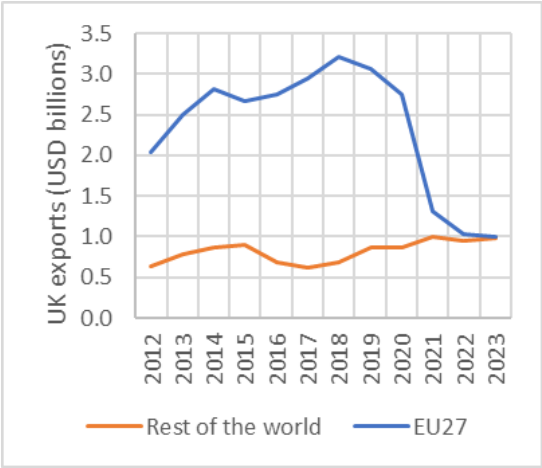

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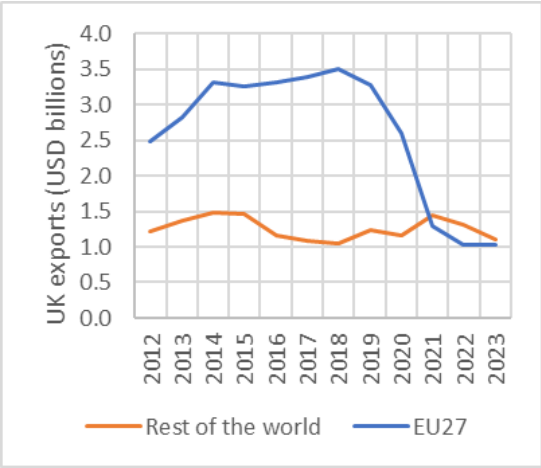

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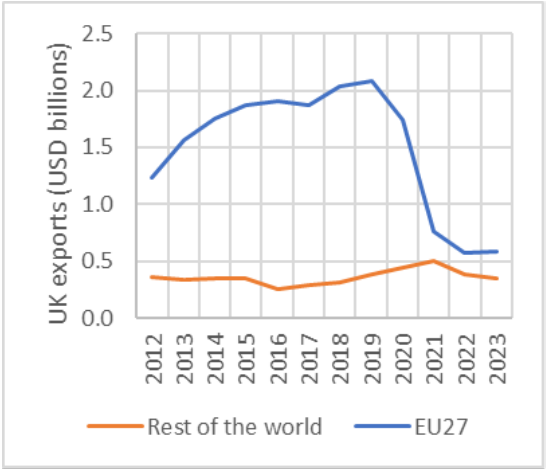

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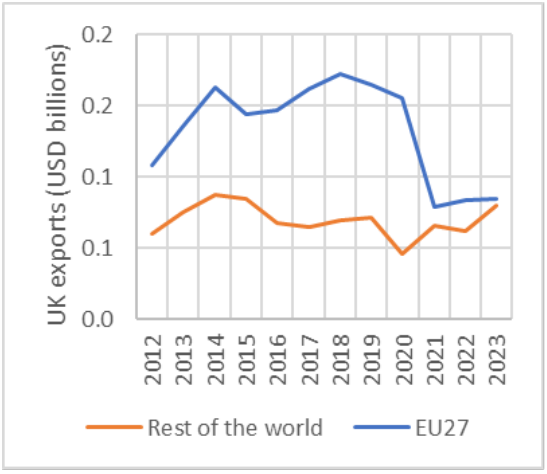

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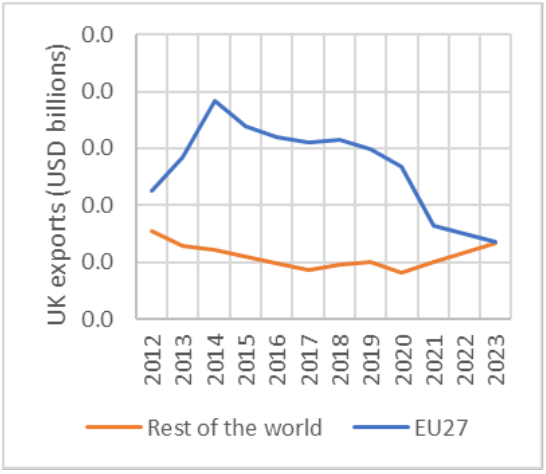

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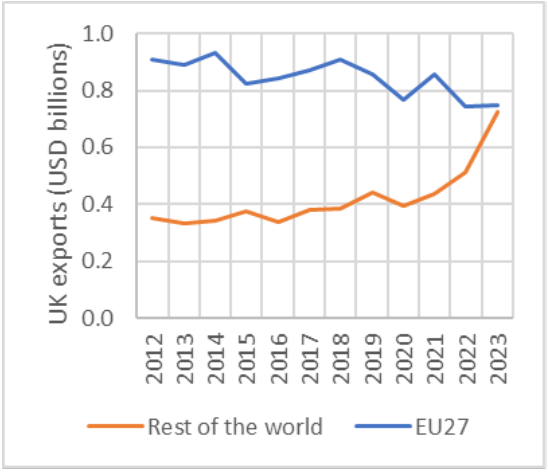

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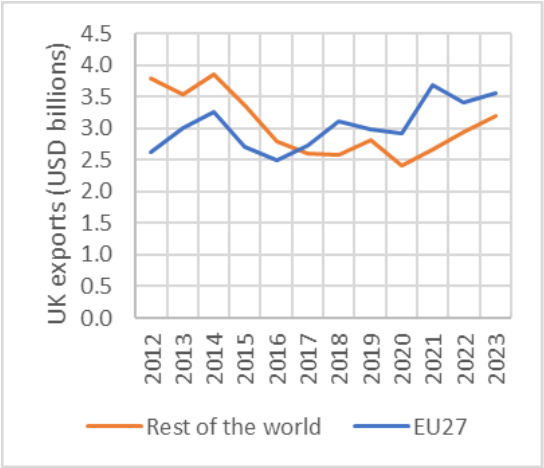

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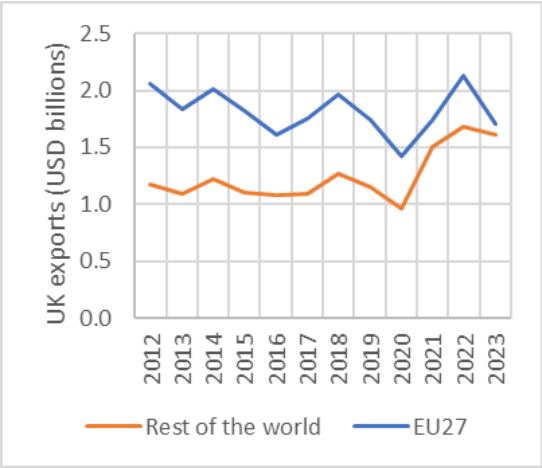

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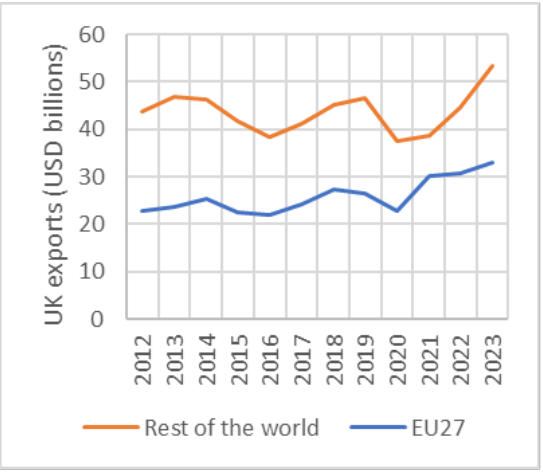

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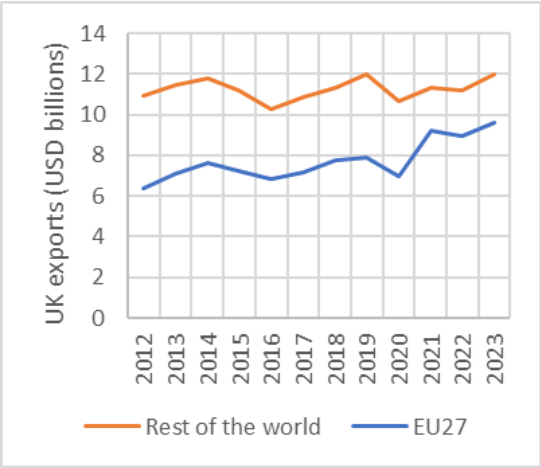

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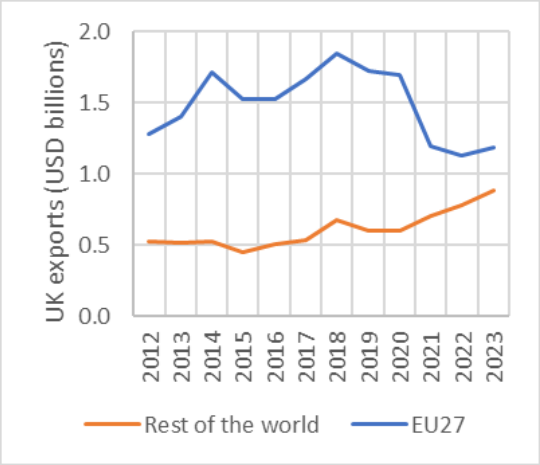

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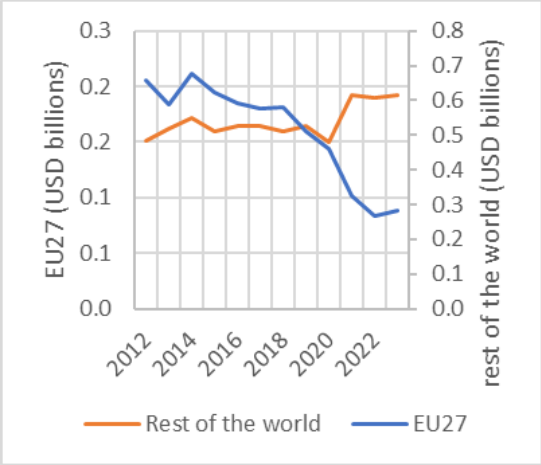

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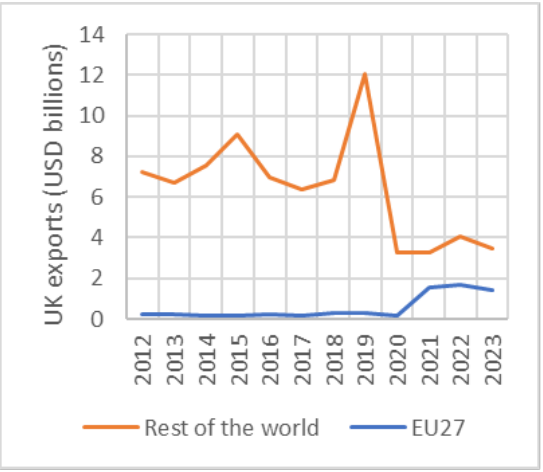

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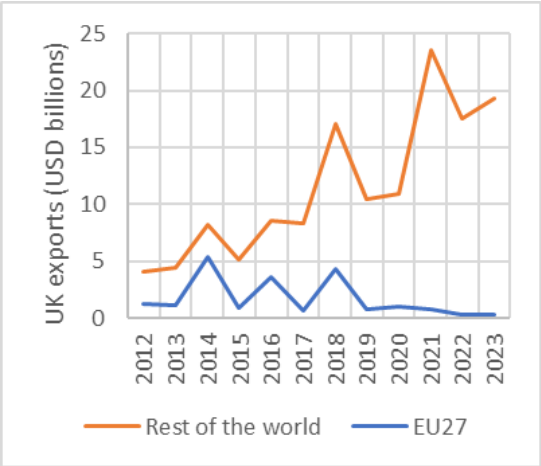

Code 99

R Code

The R code used to conduct the difference in differences analysis for the Comtrade export data is included below. The same process was used for the analysis of the import data and for analysing subcategories.

Before conducting the analysis, the code was transformed into the form seen in the table below, where region is 1 for exports to the EU27 and 0 for exports to the rest of the world.

Period	cmdCode	cmdDesc	Region	Fob Value
2012	1	Animals; live	1	3.34×10^8

```
\# Import packages
library(haven)
library(tidyverse)
library(tidyr)
library(tibble)
library(stargazer)
library(readr)
library(dslabs)
library(dplyr)
library(car)
library(plm)
library(AER)
library(ggplot2)
library(readxl)
library(fixest)

base2 <- read_excel("/Users/antoine/Downloads/Category90.xlsx")
View(base2)

base2 <- base2 %>% mutate(post2021 = ifelse(period >= 2021, 1, 0))

results_list <- list()

\# Boucle pour les cmdCode de 1    100
for (code in 1:100) {
  \# Filtrer la base pour la cat gorie en cours
  base_loop <- base2 %>% filter(cmdCode == code)

  \# V rifier si la base n'est pas vide avant d'estimer le mod le
  if (nrow(base_loop) > 0) {
    \# Estimation du mod le Diff-in-Diff
    model_loop <- lm(log(fobvalue) ~ region * post2021, data = base_loop
        )

    \# Stocker le r sum  du mod le dans la liste
    results_list[[paste0("DandD_", code)]] <- summary(model_loop)
```

```

    \# Afficher un message pour suivre la progression
    print(paste("Mod le estim  pour cmdCode =", code))
  } else {
    print(paste("Aucune donn e pour cmdCode =", code, "- passage au suivant"))
  }
}

\# Print all model summaries
print(results_list)
```

Listing 1: R code for Difference-in-Differences analysis